

SYSTEMS UNDER PRESSURE – ON-LEVEL RUBRIC

Grade 11 Algebra 2 Mathematical Modeling Investigation

INDIVIDUAL RUBRIC (6 STANDARDS)

STANDARD	4 – EXCEEDS EXPECTATIONS	3 – MEETS EXPECTATIONS	2 – APPROACHING EXPECTATIONS	1 – BELOW EXPECTATIONS
1  SEQUENCES & SERIES MODELING	- Constructs and uses recursive, explicit, arithmetic, geometric, or series-based models appropriately. - Performs accurate calculations and explains repeated or cumulative change clearly in context. - Uses equations, graphs, tables, or calculations to make meaningful predictions. - Explains where the model begins to break down or becomes unrealistic.	- Constructs and uses sequences or series appropriately. - Performs calculations accurately most of the time. - Explains repeated or cumulative change reasonably in context. - Uses mathematical models to support predictions or conclusions.	- Uses sequences or series with limited explanation or inconsistent calculations. - Predictions or interpretations may be unclear or partially developed. - Connections to the investigation may be weak.	- Little or no meaningful use of sequences or series. - Calculations, equations, or reasoning are incomplete or inaccurate. - Work is disconnected from the investigation.
2  EXPONENTIALS & LOGARITHMS	- Constructs exponential or logarithmic equations/models appropriately. - Calculates and interprets ratios, rates, thresholds, or critical values accurately. - Uses equations, graphs, and calculations to make meaningful predictions. - Explains assumptions behind exponential behavior and identifies where the model may fail.	- Uses exponential or logarithmic reasoning appropriately. - Performs calculations accurately most of the time. - Interprets growth, decay, or thresholds reasonably in context. - Uses mathematical evidence to support predictions or conclusions.	- Uses exponential or logarithmic reasoning inconsistently or with limited interpretation. - Thresholds, ratios, or calculations may be unclear or incomplete.	- Little or no meaningful use of exponential or logarithmic reasoning. - Equations, calculations, or interpretations are inaccurate or unrelated to the investigation.
3  POLYNOMIAL BEHAVIOR	- Uses polynomial equations/models appropriately to describe graph behavior. - Identifies and interprets turning points, zeros, intervals, thresholds, multiplicity, or local/global behavior accurately. - Uses equations, graphs, and calculations to explain changing system behavior. - Explains why polynomial reasoning is appropriate in context.	- Uses polynomial reasoning appropriately. - Identifies and interprets key graph features such as turning points or zeros. - Uses mathematical evidence to explain graph behavior reasonably in context.	- Uses polynomial reasoning with limited interpretation or incomplete calculations. - Connections between graph behavior and context may be weak.	- Little or no meaningful use of polynomial reasoning. - Equations, graph analysis, or interpretations are inaccurate or disconnected from the investigation.
4  MATHEMATICAL INTERPRETATION	- Clearly explains graphs, equations, calculations, and models in context. - Interprets mathematical behavior thoughtfully using accurate vocabulary. - Explains assumptions, limitations, intervals, thresholds, and predictions meaningfully. - Connects mathematics directly to the real-world system.	- Explains mathematical reasoning clearly most of the time. - Connects equations and models reasonably to the investigation context. - Uses appropriate mathematical vocabulary.	- Explanations may be incomplete, vague, or inconsistently connected to context. - Interpretation of graphs or equations may be partially developed.	- Mathematical reasoning is missing, unclear, or inaccurate. - Explanations are disconnected from the investigation.
5  REVISION + CRITIQUE	- Revises models, assumptions, predictions, or conclusions meaningfully using evidence. - Uses critique, counterexamples, or conflicting evidence thoughtfully. - Explains how understanding changed during the investigation. - Identifies limitations, uncertainty, or confounding variables clearly.	- Revises thinking using evidence or feedback appropriately. - Identifies some assumptions or limitations with reasonable explanation.	- Revision is limited or only partially supported by evidence. - Assumptions or limitations may be vague or incomplete.	- Little or no evidence of revision. - Assumptions, limitations, or critique are missing or incorrect.
6  COMMUNICATION + REFLECTION	- Communicates reasoning clearly using graphs, equations, calculations, and evidence. - Explains mathematical thinking thoughtfully in handwritten work and vlog discussions. - Reflection shows meaningful awareness of changing understanding. - Contributes consistently and meaningfully to collaborative discussion.	- Communicates reasoning clearly most of the time. - Reflection explains some changes in understanding. - Participates appropriately in discussion and investigation work.	- Communication or reflection may be inconsistent, vague, or underdeveloped. - Participation may be limited.	Communication, reflection, or participation is minimal or unclear.

GROUP RUBRIC (2 STANDARDS)

STANDARD	4 – EXCEEDS EXPECTATIONS	3 – MEETS EXPECTATIONS	2 – APPROACHING EXPECTATIONS	1 – BELOW EXPECTATIONS
1  INVESTIGATION QUALITY	- Investigation uses appropriate real-world data and mathematically meaningful models. - Equations, graphs, calculations, thresholds, intervals, and variables are interpreted clearly. - Multiple models are compared thoughtfully. - Predictions and conclusions are supported by mathematical evidence. - Confounding variables and limitations are acknowledged clearly. - Sources are credible and used responsibly.	- Investigation uses appropriate data and mathematical models. - Graphs, equations, and calculations support conclusions reasonably. - Some limitations and confounding variables are acknowledged.	- Investigation uses incomplete or partially appropriate data or models. - Interpretation may be simplistic or weakly supported. - Mathematical reasoning may be inconsistent.	- Investigation lacks meaningful mathematical modeling or evidence-based reasoning. - Conclusions are unsupported or disconnected from the data.
2  COLLABORATIVE INQUIRY	- Group collaborates consistently throughout the investigation. - Members build on, challenge, revise, and support ideas using mathematical evidence. - Shared notebook and vlogs clearly document evolving reasoning and critique. - Group uses AI tools responsibly and critically.	- Group collaborates effectively most of the time. - Members contribute appropriately to discussion, revision, and investigation work.	- Collaboration or discussion is inconsistent or uneven across members. - Revision or critique may be limited.	- Collaboration is minimal or ineffective. - Little evidence of shared inquiry or meaningful discussion.



EXPECTED OUTCOMES: Use mathematics to construct and interpret models, perform calculations, analyze key features, make predictions, evaluate limitations, and communicate reasoning about a system under pressure.